Cost of API dependency failures: evidence review

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I found no reliable source that measures the direct financial cost of semantic API changes. The strongest evidence is adjacent: outage costs, API availability, detection and recovery times, and data-quality studies. Every figure below is labelled direct or related context.

Evidence map

Direct = measures API availability, API change signalling or consumer behaviour with APIs.
Related = general IT, data or software evidence that I do not convert into an API claim.
None = nothing reliable found.

Your topic	Best evidence found	Directness
1. Financial impact of API integration failures	Outage-cost surveys (Uptime, ITIC, New Relic, Splunk) and insurer estimates for one AWS outage	Related; no API-specific dataset
2. Engineering hours debugging integrations	Postman 2019: 22.2% of API work time on debugging and manual testing; Stripe 2018: about 17 hours a week on maintenance of all code	Related
3. Cost of maintaining third-party API compatibility	Only unsourced vendor estimates	None reliable
4. Incorrect data in payments, orders, invoices, shipping	Data-quality audits (47% of new records with a critical error); Knight Capital; Mars Climate Orbiter	Related; none API-specific

Your topic	Best evidence found	Directness
5. Downtime, failed transactions, SLAs, recovery	Uptrends API uptime data; three provider postmortems; AWS S3 SLA terms. Failed-transaction counts: none found	Direct for availability
6. MTTD and MTTR	Provider postmortem timelines; New Relic, Monte Carlo and DORA benchmarks	Direct for three incidents; others related
7. Manual upgrade and migration cost	Android API lag study; small studies of migration size. No cost figures	Related
8. Silent corruption and semantic drift	LLM drift study, Cloudflare bot-score failure, Knight Capital, Mars Climate Orbiter	Qualitative; no cost data

1. Outage and downtime costs (all related context, none API-specific)

S1. Uptime Institute: 57% of major outages cost over $100,000

- **Statistic:** 57% of respondents to Uptime's 2025 annual survey said their most recent major outage cost more than $100,000, up from 54% in the 2024 survey. About one in five put it above $1 million.
- **Definition:** Cost of the respondent's most recent significant, serious or severe outage. A secondary summary says it includes direct and indirect losses; I could not confirm that in the primary text.
- **Population:** Respondents to Uptime's annual survey, mainly data-centre and digital-infrastructure operators. Sample size not stated in the press release.
- **Date:** Annual Outage Analysis 2026, published 13 May 2026; the 54% figure is from the 2025 edition, 6 May 2025.
- **Method:** Surveys plus Uptime's database of publicly reported outages. Full report is paywalled.
- **Directness:** Related. It covers data-centre and IT outages, not API changes.

- **Limits:** Self-reported cost, one outage per respondent, operator-skewed sample, and the vendor sells membership and research.
- **Source:** Business Wire, 2026, Business Wire, 2025

S2. Uptime Institute: third parties account for about two-thirds of public outages

- **Statistic:** Over nine years of tracking publicly reported outages, third-party IT and data-centre service providers (cloud, telecoms, colocation) accounted for about two-thirds.
- **Definition and population:** Outages reported in news and social media and tracked in Uptime's database; providers include cloud and internet giants.
- **Date:** May 2025 and May 2026 press releases.
- **Directness:** Related. It shows how much outage exposure sits with providers a consumer does not control; it says nothing about API changes.
- **Limits:** Public reporting is biased toward big, visible providers; "outage" definition is Uptime's.
- **Source:** Business Wire, 2026

S3. ITIC: over 90% of mid-size and large firms put an hour of downtime above $300,000

- **Statistic:** Over 90% of mid-size and large enterprises reported that one hour of downtime costs more than $300,000 on average; 41% reported $1 million to over $5 million; 97% of firms with more than 1,000 employees reported over $100,000.
- **Definition:** Each firm's own estimate of its average cost of one hour of downtime. ITIC notes that six in ten businesses cannot calculate this.
- **Population:** More than 1,000 firms worldwide.
- **Date:** Fielded November 2023 to mid-March 2024; published 2024.
- **Method:** Independent web survey, 11th annual edition.
- **Directness:** Related.
- **Limits:** Self-estimated; only firms able to estimate answer, which likely skews toward larger, more mature firms. The copy I read is hosted by a security vendor (Calyptix), so a sponsor may exist.
- **Source:** ITIC 2024 survey, Part 2

S4. New Relic: median $2 million an hour for high-impact outages

- **Statistic:** Median cost of $2 million per hour for high-impact outages, and a $76 million annual median cost of high-impact outages. Respondents with full-stack observability reported about $1 million per hour versus $2 million without.

- **Definition:** Respondent-defined "high-impact" outages in New Relic's observability forecast.
- **Population:** 1,700 IT and engineering professionals worldwide; one secondary source gives fieldwork as April to May 2025.
- **Date:** 17 September 2025. The 2024 edition reported a median 77 hours a year of high-impact downtime and up to $1.9 million an hour.
- **Directness:** Related.
- **Limits:** Commissioned by a vendor selling observability, and the headline finding is that its product halves cost, so selection and framing bias are likely. Respondents estimate costs themselves.
- **Source:** New Relic press release

S5. Splunk and Oxford Economics: $400 billion a year for the Global 2000

- **Statistic:** Downtime costs the Global 2000 about $400 billion a year, or 9% of profits.
- **Definition:** Direct plus hidden costs of unplanned downtime, where downtime is any service degradation or outage.
- **Population:** 2,000 executives from Global 2000 companies in 53 countries and 10 industries.
- **Date:** 11 June 2024.
- **Method:** Executive survey, scaled by Oxford Economics. The scaling method is in the gated report.
- **Directness:** Related.
- **Limits:** Extrapolated from self-reports, vendor-sponsored, and includes soft "hidden" costs.
- **Source:** Splunk press release

S6. Parametrix and CyberCube: estimates for the October 2025 AWS outage differ about 15 times

- **Statistic:** Parametrix estimated $500 to $650 million in losses for US companies. CyberCube's preliminary range was $38 to $581 million.
- **Definition:** Modelled financial losses from the AWS us-east-1 outage of 19 to 20 October 2025.
- **Method:** Parametrix uses its cloud monitoring data and 3,000 historical outage events; CyberCube models re/insurance exposure.
- **Date:** Parametrix 28 to 30 October 2025; CyberCube 24 October 2025.

- **Directness:** Related. It is a provider-infrastructure outage, not an API change.
- **Limits:** Both firms sell products priced on outage risk, and the gap between the two estimates shows how soft the numbers are.
- **Sources:** Parametrix, Reinsurance News

2. API availability, detection and recovery times

A1. Uptrends: average API uptime fell from 99.66% to 99.46%

- **Statistic:** Average API uptime fell from 99.66% in Q1 2024 to 99.46% in Q1 2025, which Uptrends calls 60% more downtime. Weekly downtime rose from about 34 to 55 minutes.
- **Definition:** Uptime as measured by Uptrends' synthetic monitoring checks of monitored API endpoints.
- **Population:** More than 2 billion monitoring checks across 20 industries, comparing two first quarters.
- **Date:** Published 8 July 2025.
- **Method:** Vendor's own monitoring data; the full report is gated.
- **Directness: Direct** for API availability. It does not measure change-induced failures or wrong-but-successful responses.
- **Limits:** Vendor data from its own customers' monitors, so selection bias is likely; two data points do not make a trend; synthetic checks do not test correctness.
- **Source:** Uptrends press post, State of API Reliability 2025

A2. Google Cloud, 12 June 2025: a blank-field policy record broke API serving

- **Statistic:** Site Reliability Engineering was triaging within 2 minutes, root cause identified within 10 minutes, a disable switch ready at about 25 minutes, rollout complete within about 40 minutes with recovery starting in smaller regions, and up to about 2 hours 40 minutes to fully recover us-central1. The status page window runs 10:51 to 18:18 PDT (7 hours 27 minutes). An early summary said 3 hours for the main incident.
- **Definition:** Times as stated in Google's incident report; recovery means Service Control and API serving fully recovered.
- **Directness: Direct** measurement of detection and recovery for an API platform failure. The cause was an availability bug triggered by invalid data, not a consumer-facing contract change.

- **Related fact:** Google's first incident post came about one hour after the start because its status infrastructure was also down, and some customers' monitoring running on Google Cloud failed too.
- **Limits:** One incident, self-reported by the provider. Customer losses are not stated.
- **Source:** Google Cloud incident report

A3. Cloudflare, 18 November 2025: detection in 3 minutes, root cause in about 2 hours

- **Statistic:** Impact began 11:28 UTC per the timeline table (the narrative says 11:20). The first automated test detected it at 11:31, the incident call opened at 11:35, root cause was confirmed at 13:37 (about 2 hours 9 minutes after impact), main impact resolved 14:30 (3 hours 2 minutes) and all systems normal 17:06 (5 hours 38 minutes).
- **Directness: Direct** for detection and recovery on a dependency used by many sites. Cause was a database permissions change that doubled a configuration file, not an API contract change.
- **Detail:** Cloudflare first suspected a DDoS attack, which slowed diagnosis.
- **Limits:** One incident; customer losses not stated.
- **Source:** Cloudflare postmortem

A4. AWS us-east-1, 19 to 20 October 2025: 14.5 hours end to end

- **Statistic:** The event ran from 11:48 PM PDT on 19 October to 2:20 PM PDT on 20 October (14 hours 32 minutes). DynamoDB's DNS state was identified as the source by 12:38 AM (50 minutes), all DNS information was restored by 2:25 AM, and connections recovered by 2:40 AM (2 hours 52 minutes after the start). Later phases hit EC2 launches, Network Load Balancers and other services.
- **Cause:** A latent race condition in DynamoDB's DNS automation left an empty record.
- **Directness: Direct** for provider API error rates and recovery timing; not a contract change.
- **Source:** AWS post-event summary

A5. New Relic: mean time to detect 28 minutes with full-stack observability

- **Statistic:** Respondents with full-stack observability averaged a 28-minute mean time to detect, which New Relic says is 7 minutes faster than those without (implying about 35 minutes). A separate April 2025 report gave financial services and insurance a median 42-minute detection time and 58-minute resolution time, both above all-industry figures.

- **Population:** 1,700 IT and engineering professionals (2025 forecast); sector sample size not captured.
- **Directness:** Related. It covers monitored IT outages, which are loud by definition.
- **Limits:** Vendor-commissioned; respondents report their own figures; a loud outage is much easier to detect than a silent wrong response.
- **Sources:** New Relic, September 2025, New Relic, April 2025

A6. DORA: recovery-time benchmarks for your own failed deployments

- **Statistic:** In the 2023 DORA benchmarks, elite teams restore service in under one hour, high performers within a day, medium performers between a day and a week. The 2024 clusters report says elite teams recover about 2,293 times faster than low performers.
- **Definition:** Time to recover from a failed deployment of the team's own software.
- **Directness:** Related. It is the reference point for "good" recovery, but it covers teams' own releases, not third-party changes they cannot roll back.
- **Limits:** I read secondary summaries (Apache DevLake, Octopus Deploy), not the DORA report.
- **Sources:** Apache DevLake summary of 2023 benchmarks, Octopus on the 2024 clusters

A7. AWS S3 SLA: credits of 10%, 25% or 100%, for server errors only

- **Statistic:** For S3 Standard, a monthly uptime below 99.9% earns a 10% credit, below 99.0% a 25% credit, and below 95.0% a 100% credit of the month's charges for the affected storage class. Credits apply only to future bills, must be claimed with request logs by the end of the second billing cycle, and are the sole and exclusive remedy.
- **Definition:** Uptime is 100% minus the average error rate over 5-minute intervals, where errors are "InternalError" or "ServiceUnavailable" responses.
- **Date:** Last updated 28 November 2023.
- **Directness: Direct** for SLA remedy size. **Interpretation:** because only those two error types count, a response that returns success with wrong content would not register as downtime under this definition.
- **Limits:** One provider's contract; other providers differ.
- **Source:** Amazon S3 SLA

3. Engineering effort, maintenance and migration

No source I found measures engineering hours spent on broken third-party integrations. The nearest are general maintenance surveys and academic studies of how clients lag behind API change.

E1. Stripe Developer Coefficient: about 17 hours a week on maintenance

- **Statistic:** Developers spend more than 17 hours a week on maintenance such as debugging and refactoring, including about 4 hours on "bad code"; Stripe puts the opportunity cost of bad code at about $85 billion a year worldwide and the wider technical-debt drag at about $300 billion.
- **Definition:** Self-reported time on maintenance, technical debt and bad code, valued at average developer salaries by country.
- **Population:** A survey of developers and executives. Sources disagree on the pollster (Harris Poll in two summaries, Nielsen in an Australian outlet), and I could not open the original PDF.
- **Date:** September 2018.
- **Directness:** Related. It covers all code maintenance, not third-party API work.
- **Limits:** Self-reported hours, a dollar figure built from salary assumptions, seven years old, and sponsored by an API company that benefits from a "developer time is wasted" story. Sonar's later vendor survey reported a lower average of 12 hours a week on maintenance (date not captured).
- **Sources:** i-Programmer summary, O'Reilly Radar quoting the report, Sonar

E2. Postman 2019: 22.2% of API work time on debugging and manual testing

- **Statistic:** Of time spent working with APIs, 26.1% went to development, 22.2% to debugging and manual testing and 11.4% to automated testing. About 28% said API breakages and changes occurred monthly, 15.7% weekly and 3.2% daily, while almost half said their APIs do not break or change often enough to matter.
- **Population:** More than 10,000 API developers, users, testers and executives.
- **Date:** Reported 10 December 2019.
- **Directness:** Partly direct (frequency of API breakage and change as experienced by respondents), but the question wording and whether respondents consume or provide the API were not in my source.
- **Limits:** Self-selected respondents from a vendor's community, six years old. The "almost half" finding is evidence against overstating how often breakage happens.

- **Source:** SD Times

E3. Nylas: 73% spend up to five hours a week on API maintenance and bug fixing

- **Statistic:** 73% of respondents spend "up to five hours a week" on API maintenance and bug resolution.
- **Population:** 1,200 US developers, engineering leaders and technical executives, about 60% developers. Date not captured in my source.
- **Directness:** Closest to your topic, but "up to five hours" is a ceiling, so it does not show a heavy burden and cannot support an average.
- **Limits:** Vendor survey.
- **Source:** The New Stack

E4. Android clients lag behind API change (ICSM 2013)

- **Statistic:** Android averaged 115 API updates a month. About 28% of API references in client apps were outdated with a median lag of 16 months; 22% of outdated usages eventually upgraded, taking about 14 months against a three-month release interval. Adaptation code was more defect-prone than other code.
- **Population:** Android API history and open-source client apps on GitHub.
- **Date:** 2013; ICSME 2023 named it a most-influential paper.
- **Directness:** Related. It is an operating-system SDK, not a web API.
- **Limits:** Old, open-source apps only, correlational.
- **Source:** ICSME 2023 awards page

E5. Maven Central: breaking changes are common but often harmless to clients

- **Statistic:** A 2014 study found about one third of library releases introduce at least one breaking change, equally for minor and major releases. A 2022 replication of 119,879 library upgrades and 293,817 clients found 83.4% of upgrades comply with semantic versioning, and only 7.9% of clients were affected by breaking changes, because most affect code no client uses.
- **Directness:** Related. Java libraries with compile-time checks, not web APIs with runtime behaviour.
- **Limits:** The two studies disagree on how widespread the problem is. The replication is evidence against assuming breaking changes always hurt consumers.
- **Sources:** TU Delft record, 2014, Ochoa et al. replication, 2022

E6. Web API versioning signals are unreliable (ICWE 2024)

- **Statistic:** Across 3,075 OpenAPI-described web APIs and 16,053 new versions, only 517 APIs consistently released major versions when making breaking changes, and 1,970 did not always warn clients of breaking changes in minor or patch releases.
- **Directness: Direct** for how reliably providers signal breaking change; it does not measure cost.
- **Limits:** Open-source API descriptions on GitHub; detected changes are what the specification diff shows, not behaviour.
- **Source:** Springer record

E7. Migration effort: small studies point to modest code changes, not cost

- **Statistic:** A study of 335 Python library migrations found a developer learns about 4 APIs and changes about 8 lines of code on average. In a survey of seven Java library developers, six estimated client migration effort as minor.
- **Directness:** Related, and against a dramatic reading. Neither gives hours or dollars.
- **Limits:** Code-size is not effort; the Java sample is seven people who authored the breaking changes.
- **Sources:** FSE 2024 Python migrations, Why and How Java Developers Break APIs

Not usable: unsourced vendor estimates. Pages from Truto and an open-source tool claim figures such as $30,000 to $100,000 a year per integration, $60,000 to $280,000 a year for six integrations, $5,800 per integration and four hours per breaking-change incident. None cites a study or method, and the publishers sell integration tooling. I did not use them.

4. Incorrect data and semantic mismatch

I found no study that ties incorrect data specifically to payments, orders, invoices or shipping through an API. The items below show that wrong-but-plausible data is common and that interface-meaning mismatches have produced large, documented losses in other settings.

D1. Nagle, Redman and Sammon: 47% of new data records contain a critical error

- **Statistic:** On average 47% of recently created data records had at least one critical error; only 3% of the 75 data-quality scores were acceptable (97% or higher).
- **Definition:** Friday Afternoon Measurement: managers assemble 10 to 15 critical attributes for the last 100 units of work their department completed and mark obvious errors.

- **Population:** 75 assessments from a wide range of organisations, processes and datasets, collected over two years.
- **Date:** *Business Horizons*, published 15 May 2020.
- **Directness:** Related. General enterprise data, not API-delivered data.
- **Limits:** Managers judge their own data; small non-random sample; "critical" is defined locally.
- **Source:** Harvard Business Publishing record, abstract

D2. Monte Carlo and Wakefield: 74% say business stakeholders find data problems first

- **Statistic:** 74% said business stakeholders identify data-quality issues first "all or most of the time" (47% in 2022). 68% reported an average detection time of four hours or more. Average resolution time was 15 hours per incident, up 166%. Respondents averaged 67 incidents a month, and said 31% of revenue on average was subjected to data-quality issues.
- **Definition:** "Subjected to data quality issues" means revenue touched by affected data, not revenue lost.
- **Population:** 200 data professionals surveyed in March 2023; reported as non-customers of Monte Carlo.
- **Date:** 2 May 2023.
- **Method:** Survey by Wakefield Research, commissioned by Monte Carlo, a data-observability vendor.
- **Directness:** Related. Data pipelines, not third-party API behaviour. It is the best quantified evidence I found that silent data problems are caught late and by the wrong people.
- **Limits:** Vendor-commissioned, 200 respondents, self-reported averages, 2023.
- **Source:** Monte Carlo survey page, BigDATAwire

D3. Knight Capital, 1 August 2012: over $460 million in about 45 minutes

- **Statistic:** Per the SEC, Knight's router sent more than 4 million orders to fill 212 customer orders, traded more than 397 million shares, and lost more than $460 million. The SEC imposed a $12 million penalty. Knight reported a pre-tax loss of about $440 million.
- **What changed:** A flag that formerly activated old "Power Peg" code was reused for new functionality. One of eight servers still had the old code, so the same flag value meant something different there. The SEC notes that an internal system generated 97 automated emails to personnel before the market opened.

- **Directness:** Related analogy. It was an internal deployment, not a third-party API, but it is a documented dollar loss from the same field meaning two things on different systems.
- **Limits:** One firm, 2012; the technical narrative is from the SEC order via secondary summaries.
- **Sources:** SEC press release 2013-222, SEC news digest

D4. Mars Climate Orbiter, 1999: a units mismatch with a 4.45 conversion factor

- **Statistic:** The interface specification required impulse in newton-seconds; the software produced pound-force seconds, a factor of about 4.45. The loss is put at about $125 million by a secondary source.
- **Date:** NASA Mishap Investigation Board Phase I report, 10 November 1999.
- **Directness:** Analogy only; not an API. It illustrates silent unit mismatch between two parties.
- **Sources:** Report summary, SimScale

D5. Cloudflare, 18 November 2025: a dependency returned wrong values with no errors

- **Statistic:** Customers on Cloudflare's older proxy engine saw no HTTP errors, but bot scores were not generated correctly and all traffic received a bot score of zero. Cloudflare says customers with bot-blocking rules would have seen large numbers of false positives. Customers on the new engine saw 5xx errors.
- **Directness:** Closest primary-source example I found of a provider output being silently wrong for consumers that act on it. It is a provider bug, not a contract change, and no customer losses are quantified.
- **Source:** Cloudflare postmortem

D6. Chen, Zaharia and Zou: the same model name behaves differently over months

- **Statistic:** GPT-4 identified prime versus composite numbers at 84% accuracy in March 2023 and 51% in June 2023; both GPT-3.5 and GPT-4 made more formatting mistakes in code generation in June.
- **Population and method:** March and June 2023 snapshots of OpenAI models, evaluated on benchmark tasks.
- **Date:** arXiv July 2023; *Harvard Data Science Review* 2024.
- **Directness:** Related to semantic drift behind an unchanged interface. It measures benchmark accuracy, not customer harm.

- **Limits:** I recall published critiques of the methodology, which I did not verify here. Treat the 84% to 51% figure with care.
- **Source:** arXiv 2307.09009

D7. Widely quoted figures I could not verify

- Gartner's "$12.9 million a year" for poor data quality appears in many vendor posts, one citing Gartner, July 2021. I could not open the primary report, so its definition and sample are unknown.
- "$3.1 trillion a year" attributed to IBM appears only in secondary sources and is a national extrapolation.
- A vendor rule that fixing a bad record costs $1, $10 or $100 at different stages is unsourced in what I read.

5. What is not available

These are the gaps. I did not fill any of them with an estimate.

- **Direct cost of semantic API changes:** I found no peer-reviewed study, regulator filing or provider postmortem that quantifies the financial cost of a consumer being hurt by a changed field meaning, unit, default or status value.
- **Failed transactions caused by API drift:** no counts or rates found.
- **Engineering hours on broken integrations, and cost of maintaining third-party compatibility:** no independent measurement; only general maintenance surveys and unsourced vendor figures.
- **Mean time to detect a silent change:** none found. The detection figures above are for loud outages or for data-quality incidents.
- **Why the gap exists (interpretation):** consumers rarely publish postmortems that name a provider and a loss; providers announce changes but not their effects; and silent errors are by definition not logged as incidents.
- **If your team needs a number:** the defensible route is to collect it, for example a short survey of developers about their last breaking third-party change, or a measured case study of one integration. Say plainly that it is a small sample.

Conclusions

A. The five strongest quantitative findings

1. **Version numbers do not reliably signal breaking change (E6).** Of 3,075 web APIs studied, 1,970 did not always warn clients of breaking changes in minor or patch releases, and only 517 consistently used major versions for them. Direct and peer-reviewed (ICWE 2024); measures signalling, not cost.
2. **API downtime rose 60% in a year (A1).** Average API uptime fell from 99.66% to 99.46%, or about 34 to 55 minutes a week, between Q1 2024 and Q1 2025, over more than 2 billion checks. Direct for availability; vendor data with selection bias.
3. **Outage costs cluster at six to seven figures (S1, S3, S4).** 57% of major outages cost over $100,000 (Uptime, 2025 survey); over 90% of mid-size and large firms put an hour at over $300,000 (ITIC, 2024); the median high-impact outage costs $2 million an hour (New Relic, 2025). Three different surveys agree on order of magnitude. Related context only; all self-reported.
4. **Silent data problems are found late, by the wrong people (D2).** 74% of 200 surveyed data professionals said business stakeholders find data issues first all or most of the time; average resolution was 15 hours. Related; vendor-commissioned.
5. **A field meaning two things cost over $460 million in about 45 minutes (D3).** SEC findings on Knight Capital. Related analogy: internal deployment, not a third-party API.

B. The three most important consequences for businesses

Each is my interpretation, built on the cited facts.

1. **Silent wrong output escapes the controls businesses rely on.** Availability monitoring and SLAs look for errors. The S3 SLA counts only InternalError and ServiceUnavailable responses (A7); Cloudflare's older proxy returned bot score zero for all traffic with no HTTP errors (D5); and in the Monte Carlo survey stakeholders usually find problems first (D2).
2. **Consumers carry the compatibility burden, and the recovery clock belongs to the provider.** Providers often fail to signal breaking change in version numbers (E6). Recovery after provider incidents ranged from about 40 minutes to mitigate and 2 hours 40 minutes at worst (Google, A2) to a 14.5-hour event (AWS, A4). The contractual remedy is a credit of 10% to 100% of one month's fee for the affected service (A7), against incident costs that most surveyed organisations put above $100,000 (S1).

3. **Dependency risk is large but poorly quantified.** Third-party providers account for about two-thirds of publicly reported outages (S2), yet two insurers' estimates for one AWS event differ about 15 times (S6), and no one measures the cost of change-induced failures specifically.

C. Facts suitable for a presentation slide

Fact	Suggested wording	Label to show on the slide
E6	"1,970 of 3,075 web APIs did not always flag breaking changes in minor or patch releases"	Serbout and Pautasso, ICWE 2024
A1	"Average API uptime fell from 99.66% to 99.46% in a year, about 34 to 55 minutes of downtime a week"	Uptrends, Q1 2024 vs Q1 2025, over 2 billion checks; vendor data
A2	"Google Cloud, June 2025: a policy record with blank fields crashed API serving globally; root cause in 10 minutes, longest recovery about 2 hours 40 minutes"	Google incident report
A7	"AWS S3 SLA: 10%, 25% or 100% credit of one month's charges, the sole remedy, counted only for server errors"	Amazon S3 SLA, November 2023
S1	"57% of surveyed organisations said their latest major outage cost over $100,000"	Uptime Institute 2025 survey; self-reported; IT outages, not API-specific
D2	"74% of surveyed data teams said business stakeholders usually find data issues first"	Wakefield for Monte Carlo, 200 respondents, 2023; vendor-commissioned
D3	"Knight Capital lost over $460 million in 45 minutes when a flag meant two things on different servers"	SEC 2013; internal system, not a third-party API

D. Claims that would be misleading to make

Do not say	Why
"API failures cost businesses $X an hour or $X a year" using ITIC, New Relic, Splunk, Gartner or Uptime figures	Those surveys measure IT or data-centre downtime generally. Nothing in them isolates API failures.
"$5,600 a minute" or any single per-minute downtime figure	It is a widely repeated Gartner number that I saw only in secondary pages and could not trace to a method.
"$300 billion lost to bad code" or "$85 billion" as an API cost	Stripe's figures are salary-based extrapolations about all code maintenance, from 2018.
Gartner's $12.9 million or IBM's $3.1 trillion as the cost of API drift	I could not verify definitions or samples, and neither concerns APIs.
"The AWS outage cost $500 to $650 million" as fact	It is one insurer's model; another gave $38 to $581 million. It was an infrastructure outage, not a change failure.
"31% of revenue is lost to bad data"	The survey says revenue "subjected to" data-quality issues, not lost.
"Developers spend a third of their time on API integrations"	The Stripe and Sonar figures cover all maintenance. Nylas reports up to five hours a week, a ceiling.
"Silent semantic API drift costs businesses $X"	No reliable data exists; say that this is an unmeasured gap.
"Breaking changes are rampant and hurt most clients"	The 2022 Maven replication found 83.4% of upgrades follow semantic versioning and 7.9% of clients affected. Evidence is mixed.
Knight Capital or Mars Climate Orbiter as API incidents	Neither involved a third-party API; present them as analogies.
"GPT-4 got worse, costing customers"	The paper measured benchmark accuracy only, and critiques exist.

Do not say	Why
Vendor figures such as $5,800 per integration, four hours per breaking-change incident, or $30,000 to $100,000 a year per integration	Unsourced marketing numbers.

Sources

How these were read. Opened in full: the Google Cloud incident report, Cloudflare postmortem, AWS post-event summary, AWS S3 SLA, Monte Carlo survey page and the SD Times report on Postman's 2019 survey. Everything else was read through search extracts of the pages linked, so check exact figures against the original before they go on a slide. Gated reports (Uptime, ITIC full text, Splunk, Uptrends, New Relic, DORA, Stripe's PDF, Gartner) were not opened.

Primary or near-primary (provider, regulator, institution)

- Google Cloud incident report, 12 June 2025
- Cloudflare postmortem, 18 November 2025
- AWS post-event summary, October 2025
- Amazon S3 SLA
- SEC press release 2013-222 (Knight Capital) and SEC news digest
- Uptime Institute announcements, 2025 and 2026
- New Relic, September 2025 and April 2025
- Splunk and Oxford Economics, June 2024
- Uptrends, State of API Reliability 2025 and press post
- Parametrix on the AWS outage
- Monte Carlo and Wakefield survey
- ITIC 2024 survey, Part 2

Academic

- Serbout and Pautasso, ICWE 2024
- Ochoa et al., Maven replication and Raemaekers et al., TU Delft record
- McDonnell, Ray, Kim, ICSME 2013 (awards page)
- Nagle, Redman, Sammon, Business Horizons 2020

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